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16 April 1976
SUPERSEDING
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26 February 1971

MILITARY SPECIFICATION

LINK, CARTRIDGE: METALLIC BELT, CALIBER .50, M9

This specification is approved for use by the Rock Island Arsenal, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one type of metallic belt cartridge link intended to hold caliber .50 ammunition in the form of flexible belts.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on the date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-P-116	-	Preservation-Packaging, Methods of.
MIL-W-13855	-	Weapons, Small Arms and Aircraft Armament Subsystems, General Specification For.
MIL-P-16232	-	Phosphate Coatings, Heavy, Manganese or Zinc Base (for Ferrous Metals)
MIL-I-45607	-	Inspection Equipment, Acquisition, Maintenance and Disposition of.

STANDARDS

Federal

Fed. Std. No. 356	-	Commercial Packaging of Supplies and Equipment.

Military

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-109	-	Quality Assurance Terms and Definitions.

FSC 1305

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DRAWINGS

Rock Island Arsenal
F7140393 - Link, Metallic Belt, Caliber .50, M9.

PUBLICATIONS

Rock Island Arsenal
P7140393 - Packaging Data Sheet for Link, Metallic Belt, Caliber .50, M9.
QE-G-DL 7140393 - Quality Engineering Data List for Link, Metallic Belt, Caliber .50, M9.

(Acceptance Inspection Drawings referenced in this specification form a part of QE-G-DL 7140393).

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials (ASTM)
ASTM E18 - Standard Methods of Test for Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA, 19103.)

3. REQUIREMENTS

3.1 First article. Requirements for submission of first article shall be as specified in the contract (see 6.1). Unless otherwise specified (see 6.1), the first article shall include the pilot pack (see 5.1).

3.2 Materials, construction and design. The links shall conform to the materials, construction and design requirements specified herein, on Drawing F7140393 and in MIL-W-13855.

3.2.1 Function. The links, when assembled into belts of caliber .50 ammunition and subjected to each of the following requirements, shall not break, crack, separate, deform or otherwise cause a malfunction or interference.

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3.2.1.1 Free hinging. The links, when assembled into belts, shall hinge freely and fold over without binding or kinking. Testing shall be as specified in 4.5.3.5.

3.2.1.2 Machine gun function. The belts shall operate during the firing cycle of a caliber .50, M2, heavy barrel, turret type machine gun. Testing shall be as specified in 4.4.

3.2.1.3 Twisting (fanwise). A belt of 22 rounds shall be twisted in a fanwise direction around a 6 inch diameter cylinder. When the point of the first cartridge is touching the cylinder, a pull of 5 pounds shall be sufficient to pull all the other cartridge points to within 1/32 inch of the cylinder without damaging or separating the links. Testing shall be as specified in 4.5.3.6.

3.2.1.4 Twisting (helical). A belt of 12 rounds, hung vertically and stretched by a 5 pound weight (lower 4 rounds restrained from rotation), shall be twisted 110° in a clockwise direction with a maximum torque of 15 pound-inches and 110° in a counterclockwise direction with a maximum torque of 15 pound-inches without damaging or separating the links. Testing shall be as specified in 4.5.3.7.

3.2.1.5 Walk off. A belt of 12 rounds, hung vertically and stretched by a 5 pound weight (lower 4 rounds restrained from rotation), shall withstand 100 twists without any cartridge walking off its link by more than .050 inch. One twist shall consist of 120° clockwise and return to 0°, then 120° counterclockwise and return to 0°. The links shall show no evidence of damage or separation. Testing shall be as specified in 4.5.3.8.

3.2.1.6 Extractor pull. The force required to extract the steel plug, Drawing D6516242, when the plug is inserted to the assembled cartridge depth, shall be 10 to 25 pounds. Testing shall be as specified in 4.5.3.9.

3.3 Marking. The links shall be clearly marked in accordance with Drawing F7140393 and MIL-W-13855.

3.4 Workmanship. Workmanship shall be in accordance with MIL-W-13855.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.2 Quality assurance terms and definitions. Quality assurance terms and definitions used herein are in accordance with MIL-STD-109.

4.3 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).
b. Quality conformance inspection (see 4.5).

* 4.4 First article inspection. The first article (initial production unit(s)) shall be submitted for inspection in accordance with the contract (see 6.1). The first article shall be representative of the production processes to be used during quantity production. The first article shall be subjected to the quality conformance inspection specified herein, the machine gun function test specified in 4.6.9 and such other inspection as necessary to determine that all requirements of the contract have been met.

4.5 Quality conformance inspection.

* 4.5.1 Inspection lot. Unless otherwise specified, the formation, size and presentation of inspection lots shall be in accordance with MIL-STD-105 and MIL-W-13855. Samples shall be withdrawn in accordance with MIL-STD-105 from packaged links. Inspection lots shall not exceed 500,000 links per lot.

* 4.5.1.1 Heat treat lot. The heat treat lot shall be limited to one heat of steel and shall consist of links heat treated in a batch type furnace at one charge or in a continuous type furnace during an uninterrupted period extending over not more than one eight-hour shift.

4.5.2 Examination. Examination of links and packaging shall be performed as specified herein and in MIL-W-13855.

4.5.2.1 Classification of defects. Classification of defects and acceptable quality levels (AQL's) for examination in accordance with MIL-STD-105 shall be as specified herein and shall include the following provisions:

a. Unless otherwise specified, AQL's apply to each listed characteristic.
b. Where a Government drawing number is listed as the method of inspection, the contractor shall use inspection equipment fabricated in accordance with the applicable drawing. In the event the contractor desires to use alternate equipment which equals or exceeds the accuracy requirements of the specified drawing, it shall be submitted to the Government for approval.

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c. Where Visual is specified as the method of dimensional inspection, the characteristics should be either scaled or compared with a specimen of known acceptable quality.

* d. Where Visual is specified as the method of inspection of protective coating, the coating shall be visually examined for completeness, uniformity in appearance and color, for freedom from pits, corrosion, scratches and worn or bare spots.

e. Where Standard Measuring Equipment (SME) is specified as the method of inspection, the contractor may use any type of industry developed, commercially available, multi-usage equipment or special inspection equipment approved by the Government.

4.5.2.1.1 Link, metallic belt, caliber .50, M9 (Drawing D7140393).

Class		Characteristic	AQL	Inspection Method
Critical:		None defined		
Major:				
	101	Connecting ring, outside diameter.	.65	A8409356
	102	Connecting ring, inside diameter.	.65	A8409361
*	103	Offset of bridge.	.65	D8440932
*	104	Distance between the center line of the body and neck loops and the center line of the connecting ring.	.65	D7127273
*	105	Overall length, link (over loop).	.65	SME
*	106	Distance between loop faces.	.65	SME
*	107	Length, connecting ring.	.65	SME
	108	Outside shoulder diameter, neck loop.	.65	SME
	109	Inside diameter, neck loop.	.65	SME
*	110	Length, front loop.	.65	SME
*	111	Length, rear loop.	.65	SME
*	112	Link, profile.	.25	C6057840
Minor:				
	201	Missing or defective protective finish.	2.5	Visual
	202	Missing identification. (see 3.3)	2.5	Visual
	203	Workmanship. (see 3.4)	4.0	Visual

4.5.2.1.2 Packaging. Unless otherwise specified in each listed defect, the packaging requirements are specified on Packaging Data Sheet P7140393. The inspection method shall be visual.

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Class		Characteristic	AQL
Critical:		None defined	
Major:			
	101	Improper level of preservation and packing (see procurement documents).	1.0
	102	Incorrect or illegible marking.	1.0
	103	Contaminated links.	1.5
	104	Improper containers.	1.5
	105	Container interior contaminated with dirt, lint or other foreign matter.	1.5
	106	Improperly oriented links in link cartons.	1.5
	107	Improperly oriented link cartons in shipping container.	1.5
	108	Incorrect quantity of links per shipping container.	1.5
	109	Improper closure of shipping container.	1.5
*	110	Incorrect or missing inspection lot number on each carton of links.	1.5
Minor:			
	201	Workmanship	4.0

4.5.3 Testing.

4.5.3.1 Certification. Unless otherwise specified (see 6.1), the contractor shall furnish the Government representative with certified statements that each inspection lot conforms to the materials and processes specified on the applicable drawings and specifications.

* 4.5.3.2 Protective finish testing.

* 4.5.3.2.1 Accelerated corrosion testing. Sample links shall be selected from inspection lots for each processing bath prior to application of supplementary wax coating. Lot formation and sampling shall be in accordance with MIL-P-16232, whether manual basket dip or continuous conveyor equipment is used. The sample links shall be tested for compliance with the requirements of MIL-P-16232 using the test method specified in 4.6.1.1.

* 4.5.3.2.2 Supplementary wax coating weight testing. A sample of 10 links with supplementary wax coating shall be taken from each shift's production, from each production machine and tested for supplementary wax coating weight using the test method specified in 4.6.1.2. Failure of links to meet the specified supplementary wax coating requirements shall be cause for rejection of the represented lot.

4.5.3.3 Hardness testing. Sample links shall be taken from each heat-treat lot (see 4.5.1.1) in accordance with MIL-STD-105, using

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inspection level S-4 and an AQL of .40 percent defective. The links shall be tested for hardness using the test method specified in 4.6.2.

* 4.5.3.4 Decarburization testing. A sample of 5 links shall be selected at random from each heat-treat lot (see 4.5.1.1) and tested for decarburization using the test method specified in 4.6.3. Failure of any link to meet the requirements shall cause rejection of the heat-treat lot. At the completion of this test, the links shall be scrapped.

4.5.3.5 Free hinging testing. A sample of 800 links taken at random from each inspection lot shall be tested for free hinging (see 3.2.1.1) using the test method specified in 4.6.4. Failure of any link to pass this test shall be cause for rejection of the represented lot.

* 4.5.3.6 Twisting testing (fanwise). 792 of the 800 links used in 4.5.3.5 shall be tested for twisting (fanwise), (see 3.2.1.3) using the test method specified in 4.6.5. Failure of any link to pass this test shall be cause for rejection of the represented lot.

4.5.3.7 Twisting testing (helical). A sample of 804 links taken at random from each inspection lot shall be tested for twisting (helical), (see 3.2.1.4) using the test method specified in 4.6.6. Failure of any link to pass this test shall be cause for rejection of the represented lot.

4.5.3.8 Walk off testing. A sample of 804 links taken at random from each inspection lot shall be tested for walk off (see 3.2.1.5) using the test method specified in 4.6.7. Failure of any link to pass this test shall be cause for rejection of the represented lot.

4.5.3.9 Extractor pull testing. Samples of links taken from each inspection lot in accordance with MIL-STD-105, using Inspection Level II, Code letter L, regardless of the quantity in the inspection lot, shall be tested for extractor pull (see 3.2.1.6) using the test method specified in 4.6.8. Acceptance shall be based on an AQL of .40 percent defective.

4.5.4 Inspection equipment.

4.5.4.1 Acquisition, maintenance and disposition. Unless otherwise specified (see 6.1), responsibility for acquisition, maintenance and disposition of acceptance inspection and test equipment prescribed on the Quality Engineering Data List QE-G-DL7140393, and for all other inspection equipment required to perform inspection prescribed by applicable specification, shall be in accordance with MIL-I-45607.

4.5.4.2 Accuracy of standard measuring equipment. When commercial and modified commercial inspection and test equipment is used, it must be capable of repetitive measurements to an accuracy of 10 percent of the total tolerance of the characteristic being inspected.

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4.6 Test methods.

* 4.6.1 Protective finish tests.

* 4.6.1.1 Accelerated corrosion test. Sample links shall be tested for corrosion resistance of the applied protective coating using either the salt spray (fog) test or Springfield Immersion test specified in MIL-P-16232.

* 4.6.1.2 Supplementary wax coating weight test. The sample links shall be tested by weighing the links together to the nearest milligram. The supplementary wax coating shall then be completely removed from the links by trichloroethylene or trichloroethane vapor degreasing, and the links dried and weighed. The difference between the total weights before and after removal of the supplementary wax coating divided by the number of links tested shall be considered as the weight of the supplementary wax coating of each link.

4.6.2 Hardness test. The sample links shall be tested for hardness requirements using ASTM E18 and the fixture specified on D8409387.

* 4.6.3 Decarburization test. A cross section of the sample loops (see 4.5.3.4) shall be polished, etched with a 2 to 5 percent nital solution and examined under a microscope of not less than 500 power to determine compliance with decarburization requirements of Drawing F7140393.

4.6.4 Free hinging test. The links shall be tested by assembling into 50 round belts using dummy cartridges simulating Government standard caliber .50 cartridges. The belt shall be laid out horizontally to its full length, and one end shall be drawn back and forth upon the remainder of the belt. The belt shall not bind or kink.

4.6.5 Twisting test (fanwise). The links shall be tested by assembling into 22 round belts of dummy cartridges simulating Government standard caliber .50 cartridges and using a contractor designed, Government approved test apparatus similar to that depicted in Figure 2. The first cartridge of the belt shall be loaded in the double loop of the first link. The belt shall be placed on a polished metal or glass surface with the first cartridge clamped perpendicular to the axis of a 6 inch diameter drum with the point touching the cylinder. The belt shall be drawn around the cylinder with a 5 pound weight applied to the center of the double loop of the end round, perpendicular to the axis of the cartridge. This force shall be sufficient to pull the points of all the other cartridges to within 1/32 inch of the cylinder. The links shall be examined for evidence of damage or separation.

4.6.6 Twisting test (helical). The links shall be tested by assembling into 12 round belts of dummy cartridges simulating Government standard caliber .50 cartridges and using a contractor designed, Government approved

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test apparatus similar to that depicted in Figure 1. The belt shall be hung vertically and stretched by a 5 pound weight. The belt shall be twisted 110 degrees clockwise by a torque of 15 pound-inches and 110 degrees counter-clockwise by a torque of 15 pound-inches while the lower 4 rounds are restrained from rotation. The links shall be examined for evidence of damage or separation.

* 4.6.7 Walk off test. The links shall be tested by assembling into a 12 round belt using dummy cartridges simulating Government standard caliber .50 cartridges. Use the same test apparatus as the helical twisting test. Measure and record the original position of the cartridges. Twist the belt 100 times through the specified angle and measure the position of the cartridges after twisting. The difference between this position and the original position of the cartridges shall be recorded as cartridge walk off. The links shall be examined for evidence of damage or separation.

4.6.8 Extractor pull test. The links shall be tested using the steel plug and test equipment specified on D6516242. The link shall be placed in the test fixture, the steel plug inserted, and a force gradually applied to extract the plug from the link. The force required to extract the plug from each link tested shall be recorded as extractor pull.

4.6.9 Machine gun function test. The sample links shall be tested by assembling with Government standard caliber .50 cartridges into four belts of 100 linked cartridges and fired in a caliber .50, M2, heavy barrel, turret type machine gun. Two belts shall be fed from the right side feed and two belts fed from the left side feed.

5. PACKAGING

5.1 Pilot pack. A pilot pack shall consist of links preserved in accordance with Packaging Data Sheet P7140393 for the level of protection specified in the contract (see 6.1), packed level B and forwarded as specified in 3.1.

5.2 Levels A and B. Preservation, packing and marking shall be in accordance with Packaging Data Sheet P7140393 for the level of protection specified in the contract (see 6.1).

5.3 Commercial packaging. Preservation, packing and marking shall be in accordance with Fed. Std. No. 356.

6. NOTES

6.1 Ordering data. Procurement documents should specify:

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6.1.1 Procurement requirements:

a. Title, number and date of this specification.
b. List of drawings, specifications and publications pertinent to the link showing applicable revision dates.
c. Requirements for submission of first article and pilot pack (see 3.1, 4.4 and 5.1).
d. Certification, when not required (see 4.5.3.1).
e. Responsibilities for acquisition, maintenance and disposition of acceptance inspection equipment, if different (see 4.5.4.1).
f. Selection of applicable level of protection (A or B), or commercial packaging (see 5.1, 5.2 and 5.3).

6.1.2 Contract data requirements. When warranted, requirements for reports of the results of examination and testing shall be specified on a DD Form 1423 included in the contract.

6.2 Unless otherwise specified (see 6.1), the contract should specify the application of MIL-I-45607 on the Management Control Summary List, DD Form 1660.

6.3 When warranted, the contract should specify the application of MIL-Q-9858 or MIL-I-45208, as appropriate, on the Management Control Systems Summary List, DD Form 1660.

6.4 Supersession data. This specification includes the requirements of Purchase Description RIAPD-572A, dated 24 July 1967.

6.5 The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on its entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodian:
Army - WC

Review activity:
Army - MU

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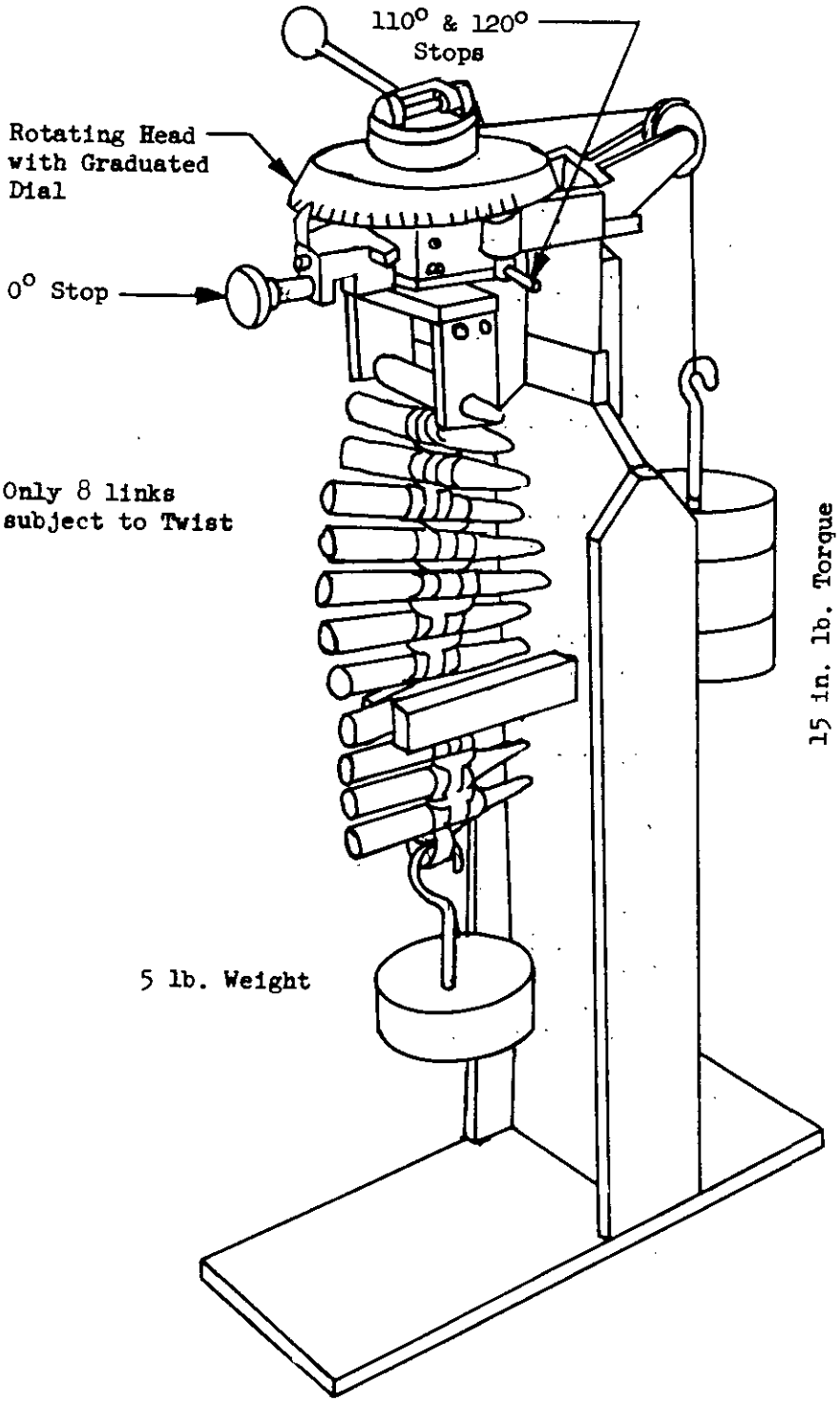


FIGURE 1. Strained helical flexibility and walkoff tester for cal .50, M9, metallic belt link.

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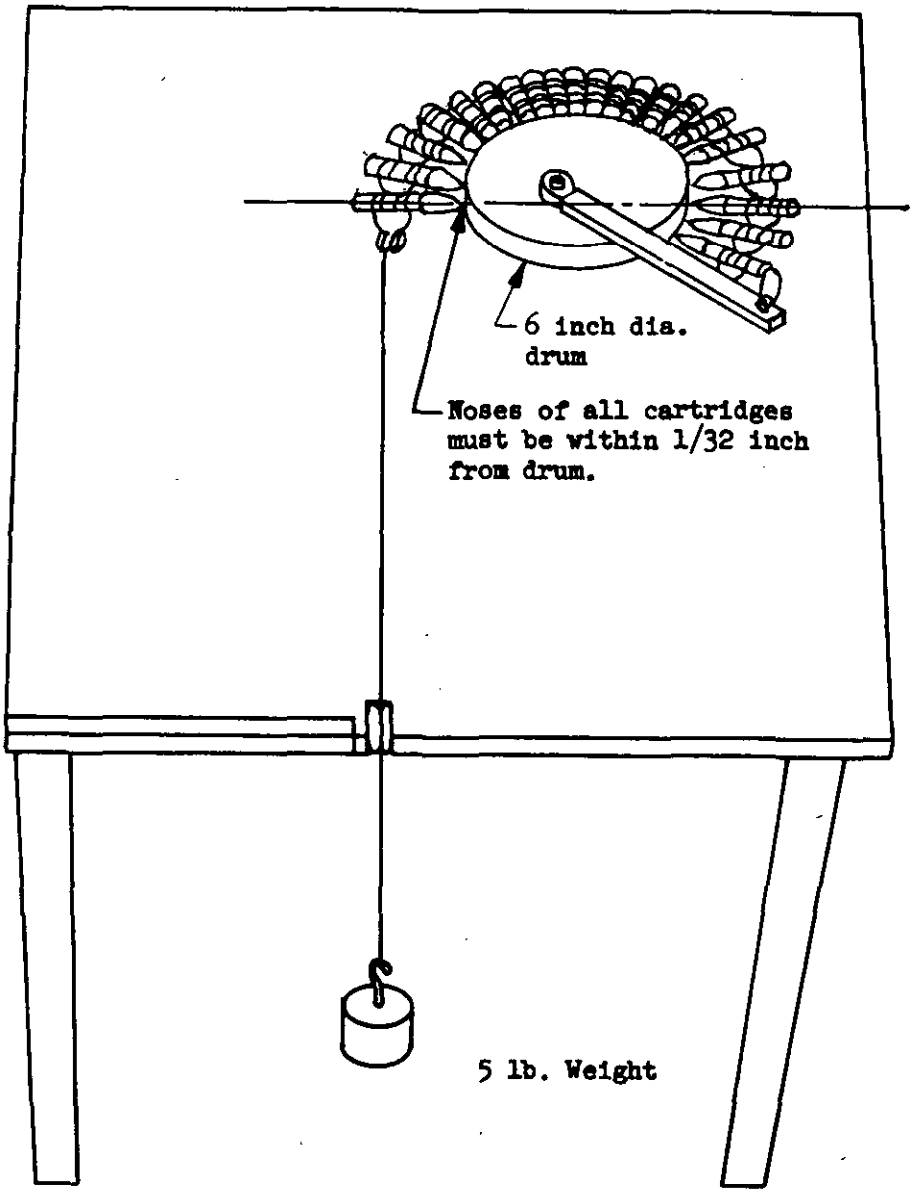


FIGURE 2. Strained fanwise flexibility tester for cal .50, M9, metallic belt link.

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